TAO PeopleNet Transformer person detection

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> Applicable environment: Jetson Orin NX, JetPack 7.2.1, DeepStream 9.1. The video entry point can run and preview.

1. Case objective and processing flow

The PeopleNet Transformer of this course configuration filters out the other classes and **only outputs person**; it does not detect face or bag. It uses different model assets and a different configuration from the CNN version of PeopleNet, and the engine/parser must not be mixed between them.

```
Video/USB/CSI → Transformer TensorRT inference → detection box/class → desktop preview
```

2. Checks before running

```
cd ~/deepstream_tao_apps/deepstream_app_tao_configs

# Confirm the configuration and the input source
grep -E "uri=|config-file=|custom-lib-path|parse-bbox" run_transformer.txt

# Confirm the dedicated Transformer model and the engine
ls -la /opt/nvidia/deepstream/deepstream-
9.1/samples/models/tao_pretrained/peoplenet_transformer_vdeployable_v1.1/

# Confirm the custom parser
ls -la ~/deepstream_tao_apps/post_processor/libnvds_infercustomparser_tao.so
nm -D ~/deepstream_tao_apps/post_processor/libnvds_infercustomparser_tao.so | grep DDETR
```

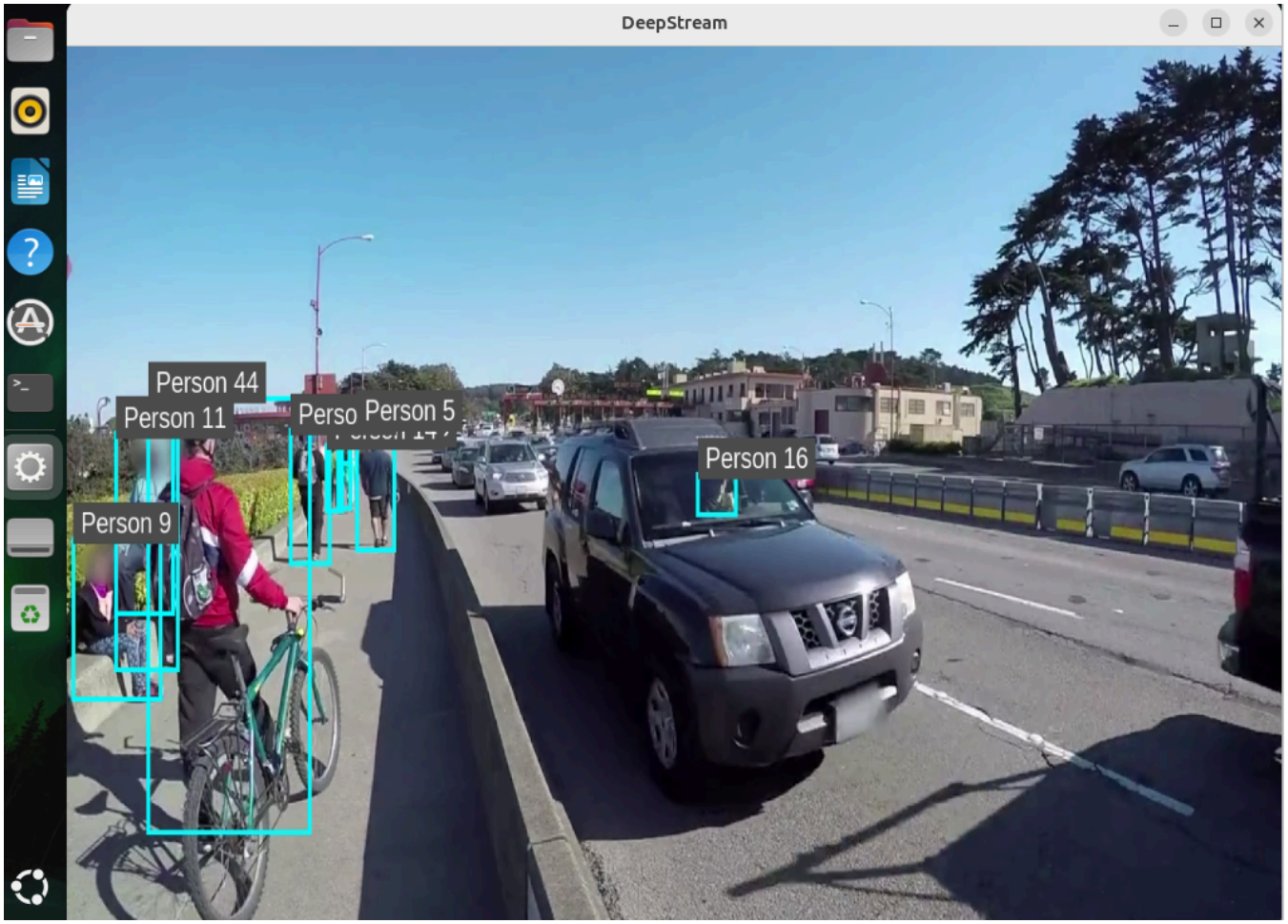
Please operate in the graphical session of the Jetson desktop/VNC. Confirm that the input exists; the Transformer has a high computational load, so the offline frame rate may be lower than the frame rate of the material.

3. Starting the detection

Video mode:

```
cd ~
export DISPLAY=:0

export DISPLAY=:0 && ~/deepstream_tao_demo/run_case.sh transformer video
```



The entry point uses the `sample_720p.h264` that comes with DeepStream, and the window shows the detection results related to people. The current configuration by default only previews or outputs to the internal sink.

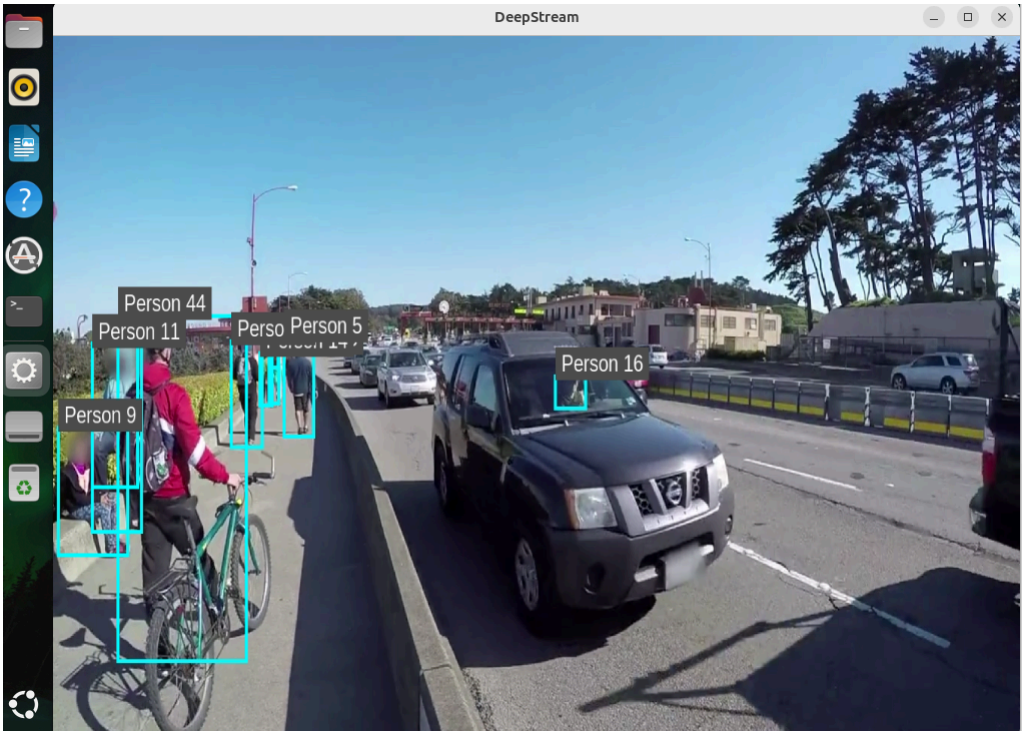
To replace the test video: modify the `uri=file://` of the `[source0]` section in `run_transformer.txt` to the absolute path of an MP4/H.264 file on the Jetson itself. First use `gst-discoverer-1.0` to confirm that the device can decode it, and keep the original material.

4. Real-time detection with a USB camera

After confirming that the USB camera device node exists, run:

```
ls -l /dev/video*
v4l2-ctl -d /dev/video0 --list-formats-ext

export DISPLAY=:0 && ~/deepstream_tao_demo/run_case.sh transformer usb
```



The script automatically detects the USB camera node and the supported resolution/frame rate.

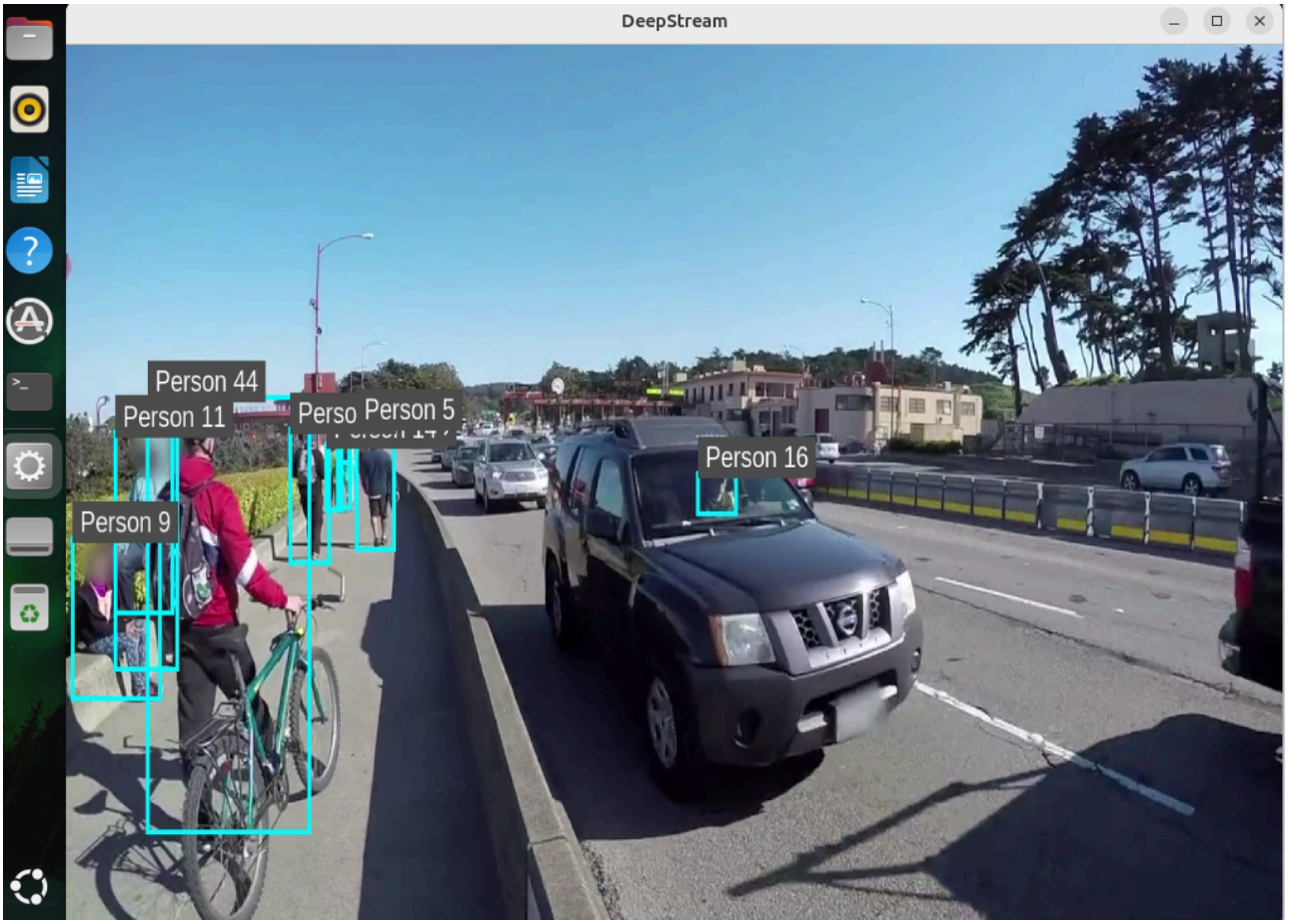
The USB mode is used for the real-time preview. The script preferentially generates a temporary `640x480@15` configuration that the camera supports; the frame only shows `Person + Tracker ID` and does not show face/bag.

5. Real-time detection with a CSI camera

The CSI mode supports the real-time preview, and the entry point captures through Argus.

Start:

```
export DISPLAY=:0 && ~/deepstream_tao_demo/run_case.sh transformer csi
```



USB/CSI is a continuous stream and does not naturally play to the end. The frame only shows person and the Tracker ID. To stop it, press `Ctrl+C` once in the terminal where it was started, wait for the application and the camera to be released, and only then start another camera task.

6. Common issues

Symptom	Check and handling
No window is shown	Check the graphical session of the Jetson, the input file and the display authorization.
The class/the result is abnormal	Check the dedicated Transformer engine, the parser and the labels; do not use the CNN PeopleNet assets.
It reports that `NvDsInferParseCustomDDETRTAO` cannot be found	Confirm that the function is in `~/deepstream_tao_apps/post_processor/libnvds_infercustomparser_tao.so`: `nm -D ... | grep DDETR`.
`Failed to load config file`	Check the `config-file=` path in `run_transformer_csi.txt` (the trap of the path missing the `configs/` prefix was hit before).
The processing frame rate is low	The Transformer load is relatively large; it was measured at about 20 FPS on an Orin NX.
`ONNX file does not exist`	Check that the ONNX path referenced by the `config-file` is valid; the `resnet50_peoplenet_transformer_op17.onnx` under the `peoplenet_transformer_vdeployable_v1.1` directory comes with DS 9.1.
No picture with USB/CSI	Check the device node, the format, the permissions and the occupancy; the CSI entry point directly uses the Argus `type=5`.
`nvv4l2h264enc has no property named 'gpu-id'`	It is a known non-blocking warning of DS 9.1 and does not affect the inference or the display.

7. References

- NVIDIA DeepStream TAO application examples: https://github.com/NVIDIA/DeepStream/tree/main/src/a

pps/tao_apps

- DeepStream TAO integration documentation: https://docs.nvidia.com/metropolis/deepstream/dev-guide/text/DS_TAO_integration.html